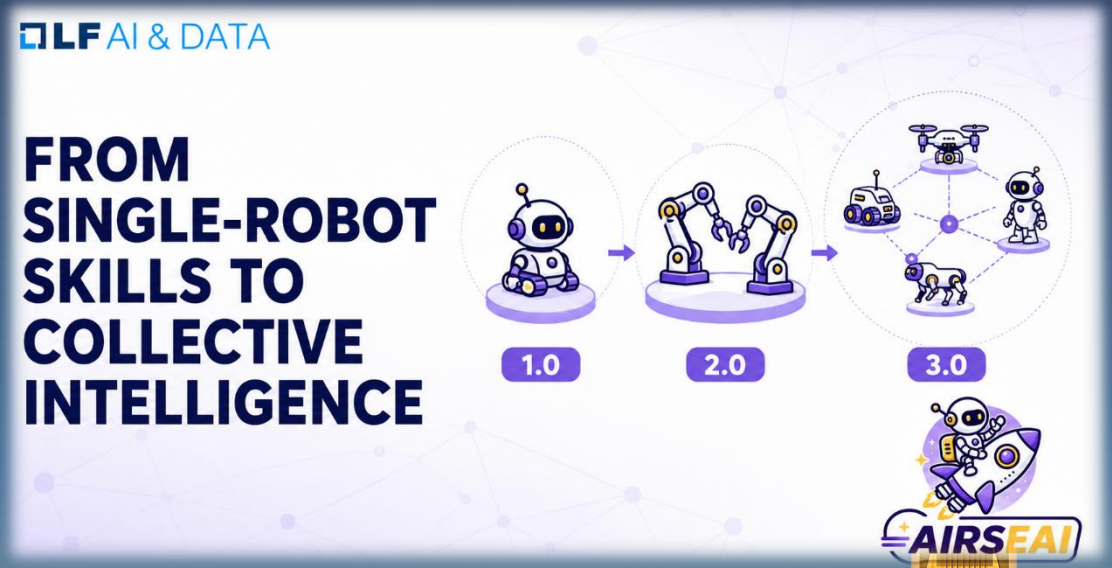
GOSIM

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AIRSEAI

Open-Source Embodied AI Ecosystem



Shenzhen Institute of Artificial Intelligence and Robotics for Society (AIRS)

深圳市人工智能与机器人研究院

周鸿钧（Hongjun Zhou）

康留旺（Liuwang Kang）

CONTENTS

Part 01 AIRSEAI 1.0 + AIRSEAI 2.0

Part 02 AIRSEAI 3.0



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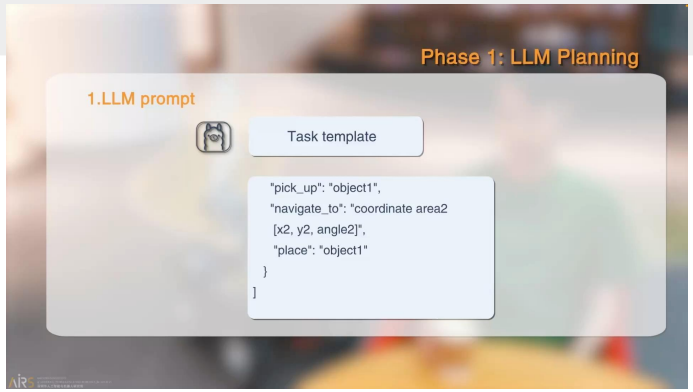
AIRSEAI 1.0 + AIRSEAI 2.0



System Evolution

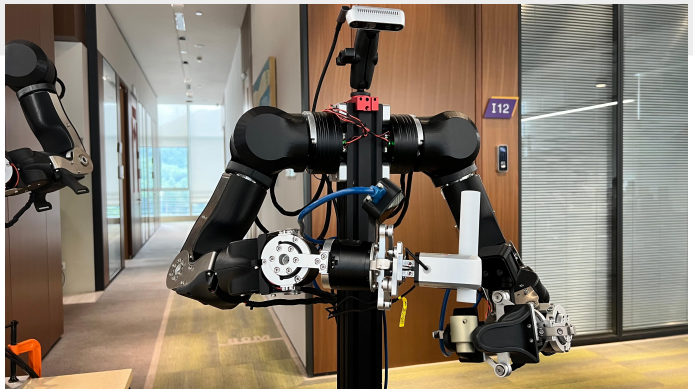
AIRSEAI 1.0

Focuses on LLM-driven high-level task planning and modular navigation using a single-arm mobile chassis.



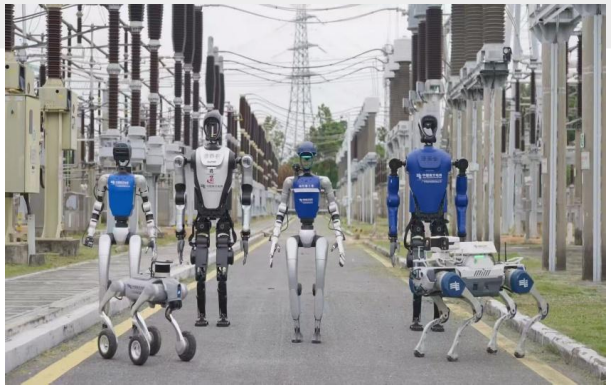
AIRSEAI 2.0

Evolves into a dual-arm general-purpose manipulation platform powered by a standardized data-driven pipeline.

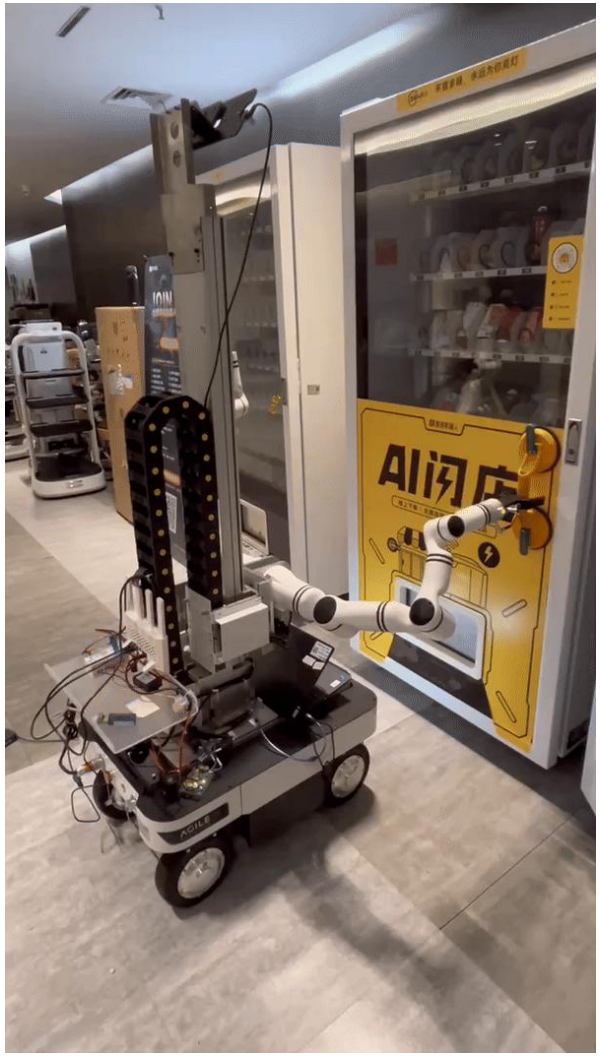


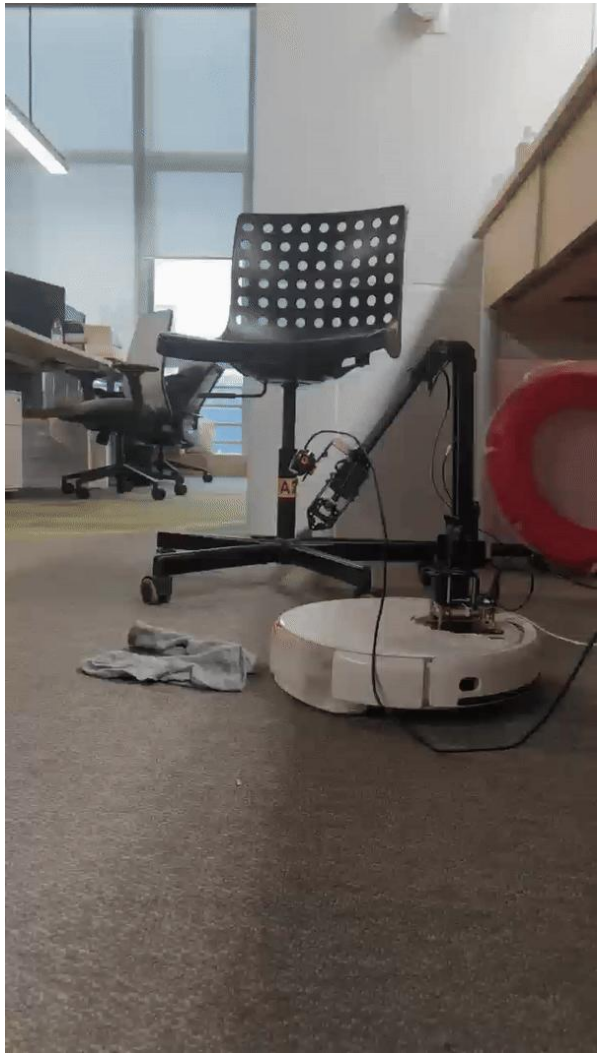
AIRSEAI 3.0

Addresses how multiple agents achieve system-level coordination through task allocation, adaptive planning, and communication-aware collaboration



AIRSEAI 1.0 Ecosystem





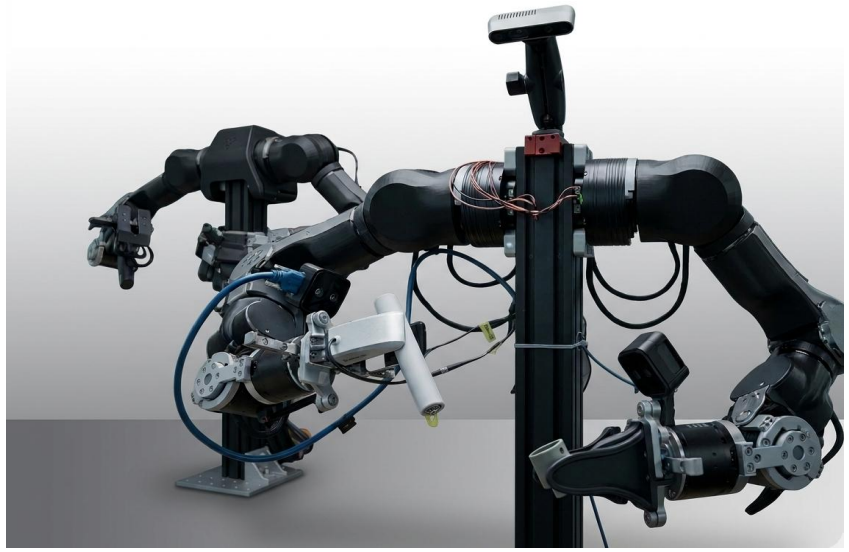




AIRSEAI 1.0



AIRSEAI 2.0



From Single to Dual-Arm

AIRSEAI 2.0 introduces the OPENARMS dual-arm system, significantly expanding the workspace and capability for complex bimanual tasks.

AIRSEAI 2.0: Specifications

Subsystem	Component Description
Robotic Platform	OPENARMS dual-arm system with leader-follower data collection
Vision System	Intel RealSense D435 (Head) + Dual D405 (Wrist-mounted)
End Effector	**Two-finger electric gripper** for 7-Dof manipulation
Tactile Sensing	**Integrated vision-based tactile sensors** in the fingertips
Computing	NVIDIA Jetson AGX Orin & RTX high-performance platforms

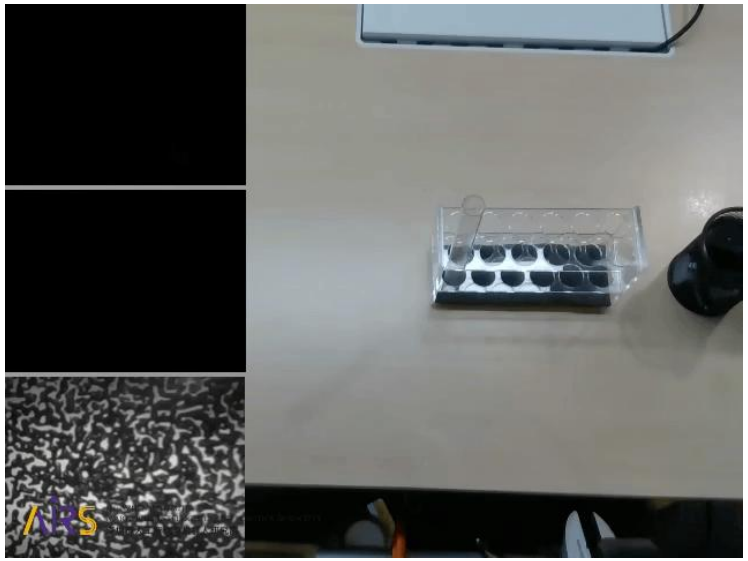


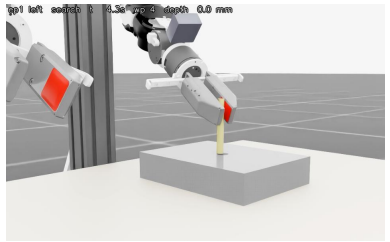
High-Resolution Tactile Perception

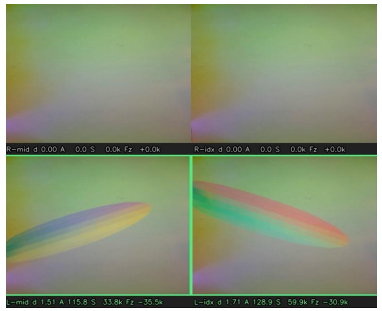
Vision-Based Tactile Sensors

The transition from resistive/capacitive sensing to vision-based tactile sensors allows for high spatial resolution in contact sensing.

By capturing the deformation of an internal elastomer with a mini-camera, the robot "sees" what it touches.







AIRSEAI 2.0

An open-source embodied-AI platform: a dual-arm manipulator with vision-based tactile sensors, one data pipeline, and two places it runs — sim and the real robot.

In simulation, this repository. A bimanual OpenArm with four tactile fingers in Isaac Sim, a scripted expert per task, LeRobot v3 datasets carrying the tactile channels, ACT training on them, and policy rollout in the same scenes.

On the real robot. The same arms, the same recorded keys, collected by leader-follower teleoperation. Sim and real do not line up everywhere, and quantifying where they part is why the benchmark exists.

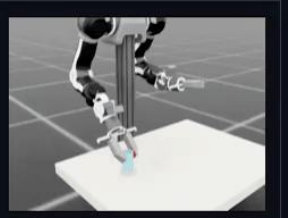
The tasks



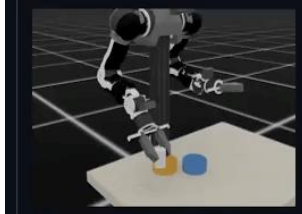
`stack_cube` — stack one cube on another



`blind_peg_insertion` — insert a peg into a hole that is not where the robot is told



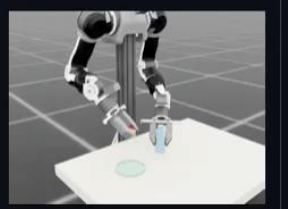
`sleeve_on_shaft` — slide a squeezable sleeve onto a shaft



`texture_sort` — sort two balls that differ only in surface, in the dark



`handover` — hand a spring-loaded sleeve between the arms, from one cup to the other



`cap_unscrew` — hold a bottle, unscrew its cap and place it in a tray

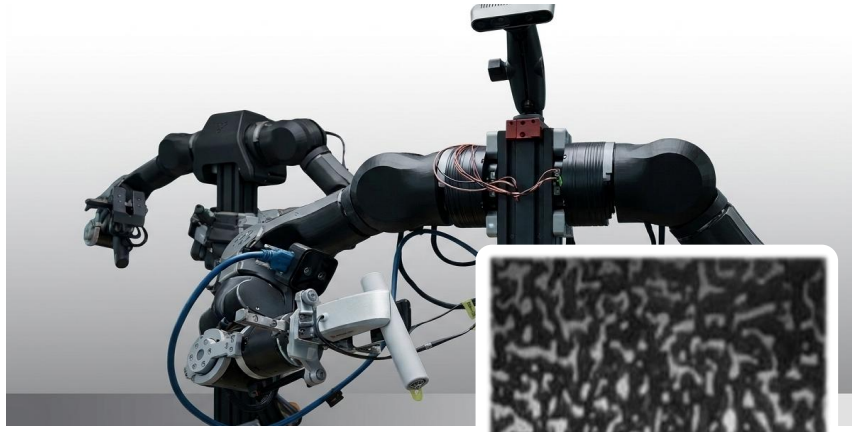
1



Collect demonstration data

VR-based teleop • vision + tactile streams

Multimodal LeRobot datasets



Fingertip tactile view

OPENARMS reference hardware
or any dual-arm platform

VR-based data collection
immersive teleop demonstrations

Extended LeRobot format for tactile
video encoding for large tactile streams

Benchmark
capability boundary tests

2

ACT | Diffusion policy

VLA • π, GR00T | + Your model

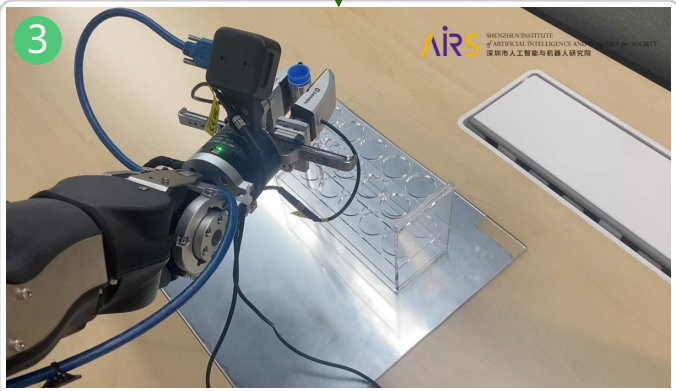
LeRobot engine • community

Train & fine-tune with LeRobot

Bring any policy - ACT, diffusion, VLA

Fine-tuned policies

3



Deploy on real hardware

Fine-tuned policies run bimanual tasks

Evaluation results

Analysis feedback

4

Evaluate on the sim benchmark

Capability boundary

AIRSEAI 2.0 Demo: Dual-Arm Manipulation







What Stayed: Continuity from AIRSEAI 1.0 to 2.0

Modular Interface

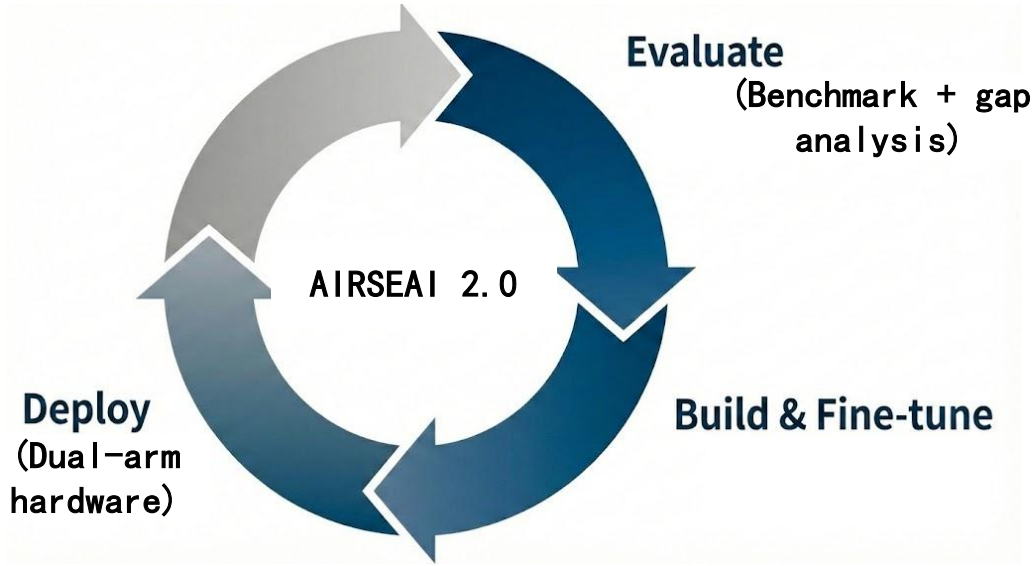
- The plug-and-play modular design philosophy from 1.0 is the direct foundation for 2.0's hardware abstraction layer.
- Standardized interfaces ensure components remain swappable.

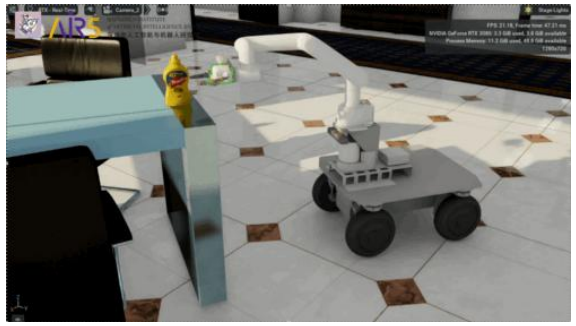
Sim-to-Real Co-Design

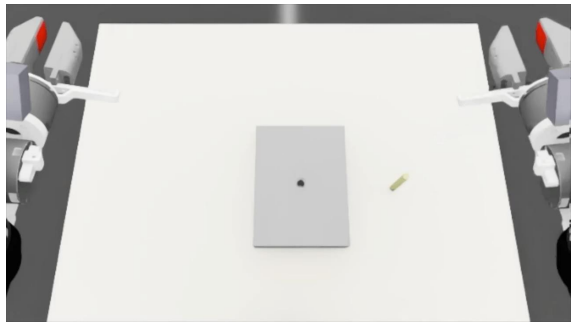
- 1.0 pioneered Isaac Sim-based validation.
- 2.0 extends this with formal gap analysis to quantify and close the simulation-to-reality performance gap.

Open Ecosystem Stance

- Open-source codebase and community-driven approach remain central. 2.0 adds hardware-agnostic compatibility with third-party platforms.



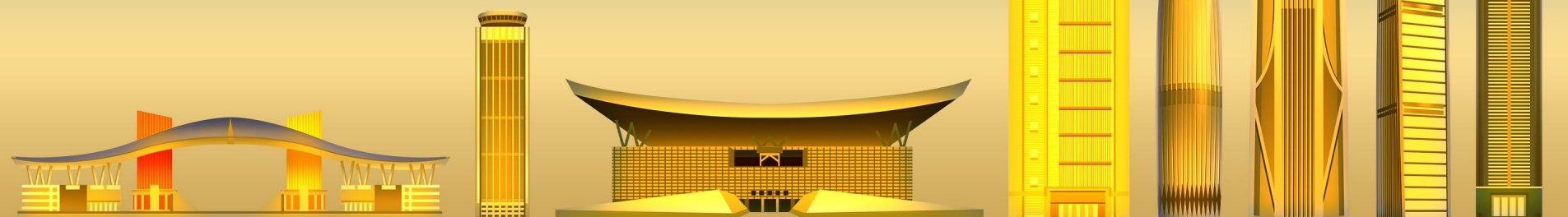




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AIRSEAI 3.0



Toward Multi-Agent Task Allocation, Adaptive Planning, and Communication-Aware Collaboration

- Multi-Robot Task Allocation and Adaptive Planning
- Communication-Aware Collaboration
- Multi-Agent Training Platform and Collective Adaptation



2025 World Internet of Things Expo
Industry Pioneer Award

Why Multi-Robot Scenarios in AIRSEAI 3.0?

Real-world tasks often cannot be solved simply by making a single robot more capable.



Limited Spatial Coverage

One robot can only observe and operate within a limited area.

Limited Parallelism

Multiple distributed tasks cannot be efficiently handled at the same time.

Single-Point Failure Risk

A robot failure may directly interrupt task execution and reduce system continuity.

Multi-robot systems extend coverage, parallelism, and system resilience.

Challenges in Multi-Robot Coordination

Adding more robots increases system capability but also introduces new coordination challenges.

Dynamic Tasks

Tasks, priorities, and resources may change during execution.

Coordination Coupling

One robot's action may change the feasible choices and roles of other robots.



Heterogeneous Capabilities

Robots differ in mobility, sensing, manipulation, energy, and communication.

Communication Constraints

Weak links, blockage, and disconnection can disrupt teaming and scheduling.

Multi-robot coordination must handle task dynamics, capability differences, coupled decisions, and imperfect communication.

Four Questions for Multi-Robot Coordination

The coordination problem can be organized around four recurring questions during mission execution.

01 WHO?

Who should do it?

Multi-Robot Task Allocation

Assign robots and roles

02 HOW?

How should it be done?

Role Coordination & Path / Action Planning

Coordinate roles and actions

03 CHANGE?

What if conditions change?

Dynamic Reorganization & Adaptive Planning

Reorganize when conditions change

04 DISCONNECTED?

What if communication is lost?

Communication-Aware Collaboration

Maintain collaboration under weak links

These four questions define the system capabilities AIRSEAI 3.0 must support.

Multi-Robot Coordination Capability Framework

Organizing coordination into four implementable and measurable system capabilities.

01

Task Allocation

- Task discovery
- Capability matching
- Dynamic teaming

02

Adaptive Planning

- Replanning after task changes
- Reorganization after robot failures
- Role switching

03

Communication-Aware Collaboration

- State/task semantic exchange
- Message relaying under weak communication
- Graceful degradation under local disconnection

04

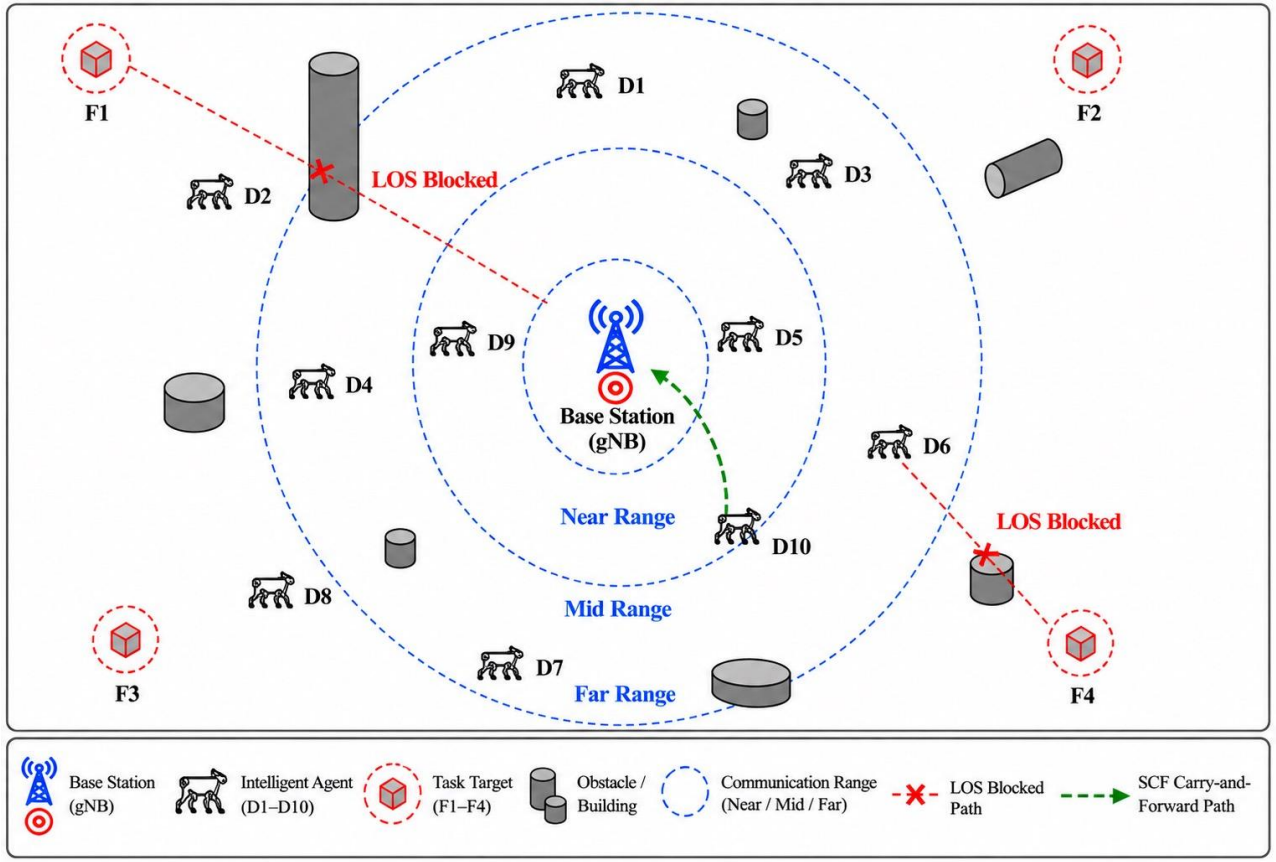
Agent Training

- Multi-robot trajectory experience
- In-context learning / policy distillation
- Simulation randomization

Fan Wang and Shaoshan Liu. "Environmental Embodied AI", Process of the ACM, 2025
Liuwang Kang, Shaopeng Tao, Fan Wang, and Shaoshan Liu. "Collective Adaptive Intelligence as a Plane for 6G Networks". IEEE Communications Magazine, 2026.

Multi-Robot Validation Scenario

A measurable scenario integrating task allocation, adaptive planning, and communication-aware collaboration



It captures the key challenges of multi-robot coordination:

Task Allocation

Which robots should transport the resource? How many partners are needed?

Adaptive Planning

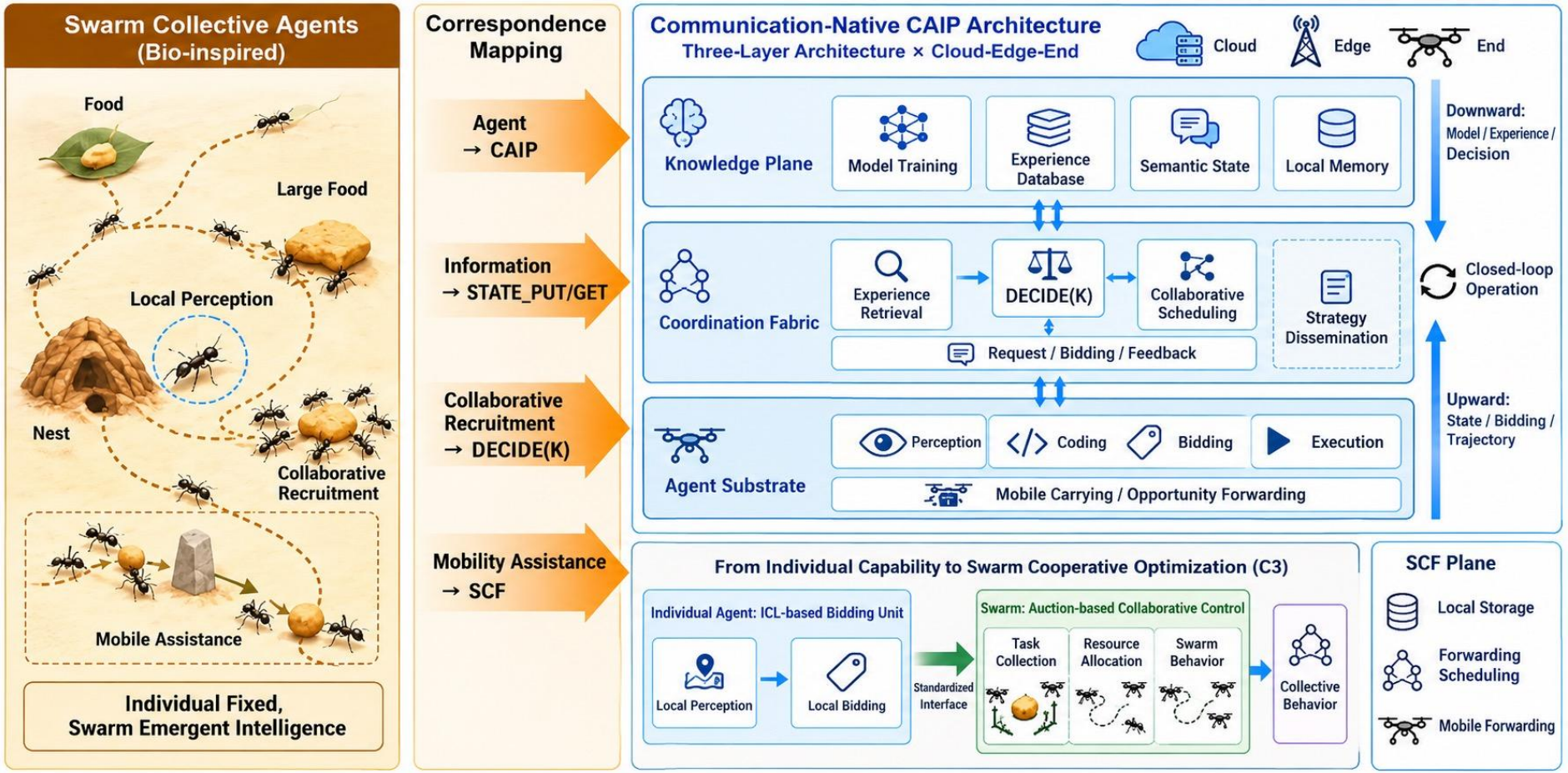
How should teams be reorganized when resources change or robots become occupied or fail?

Communication-Aware Collaboration

How can collaboration continue when obstacles and weak networks prevent message delivery?

Ant-Inspired Multi-Robot Coordination

Inspired by ant recruitment and mobile relaying, design a collectively adaptive intelligence as platform (CAIP) to enable robot collaboration in dynamic, weak-communication environments.



Liuwang Kang, Shaopeng Tao, Fan Wang, and Shaoshan Liu. "Intelligent Plane Design and Mechanism Research for Ordnance Collaborative Systems." Fire Control & Command Control, 2026.

CAIP Maps to Four AIRSEAI 3.0 Capabilities

Task allocation and adaptive planning translate changing task conditions into dynamic team formation and reorganization.

01 Task Allocation

Who should do it?

Task Discovery → State Assessment → ICL Bidding → Team Formation

- Integrate task requirements, robot states, and local experience.
- Dynamically select participating robots and assign roles.

Outcome: the right robots form the right team for the current task.

02 Adaptive Planning

What if conditions change?

State Update → Bounded Decision → Role Adjustment → Team Reorganization

- Adapt dynamically to changes in tasks, robot states, or coordination conditions.
- Move from action replanning to collaboration replanning.

Outcome: the team structure adapts as mission conditions evolve.

CAIP turns task changes into coordinated changes in team membership, roles, and execution plans.

CAIP Maps to Four AIRSEAI 3.0 Capabilities

Communication-aware collaboration and experience learning keep the coordination loop effective under weak links and evolving missions.

03 Communication-Aware Collaboration

What if communication is lost?

Semantic Sharing › Link Awareness › Local Execution › SCF Store-Carry-Forward

- Adapt collaboration strategies to current communication conditions.
- Use robot mobility to sustain information propagation under weak communication.

Outcome: collaboration degrades gracefully instead of collapsing when links fail.

04 Agent Training & Experience Learning

How are capabilities continuously developed?

Models / Experience › Experience Retrieval › ICL Reasoning › Execution Feedback

- Historical experience and current context jointly support on-device decisions.
- A cloud–edge–device loop accumulates and updates coordination experience.

Outcome: execution feedback becomes reusable experience for future coordination.

Together, task allocation, planning, communication, and learning form a closed loop for multi-robot coordination.

Experimental Setup: Testing Multi-Robot Resilience

Stress-testing across three dimensions: communication degradation, obstacle blockage, and scale variation

Simulation Platform

ns-3.38 + 5G-LENA v2.4
Simulating realistic 5G/6G link conditions

Baselines

Greedy | Centralized Rules | CAIP

Statistical Design

25 seeds | Paired experiments | 95% confidence intervals

Stress-Test Matrix

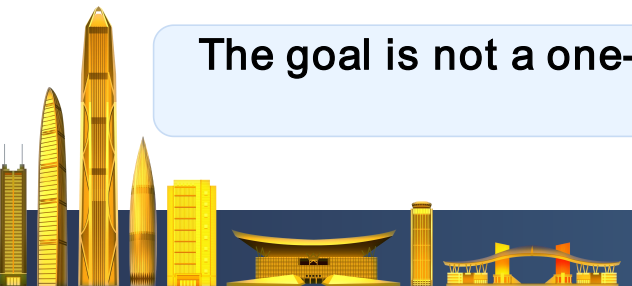
Normal | Degraded | Failed × No Obstacles | With Obstacles + 6–24 Agents

Core Metrics

Task Delivery Rate

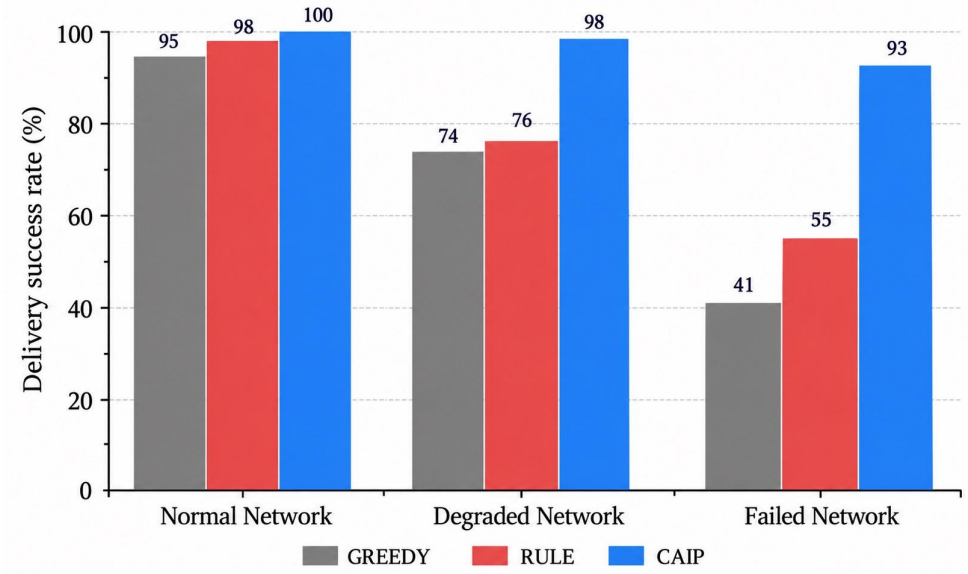
Mean Execution Steps

The goal is not a one-off successful demo, but reliable coordination under weak communication, blockage, and scale changes.



Validation I: Resilience Under Weak Networks

With obstacles, CAIP maintains delivery performance more effectively as network conditions degrade.



Resource Delivery Rate with Obstacles (10 Agents, 25 Seeds, %)

Conclusion:

- The weaker the network, the greater CAIP's relative value.
- CAIP is not designed to create a gap under perfect connectivity, but to prevent coordination collapse when communication fails.

Network Failure: CAIP



Network Failure: Greedy



Network Failure: Rules

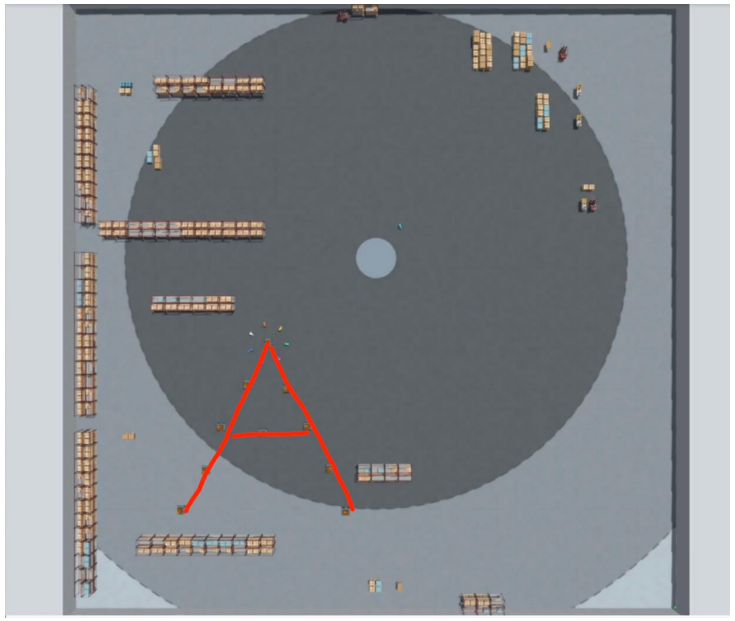


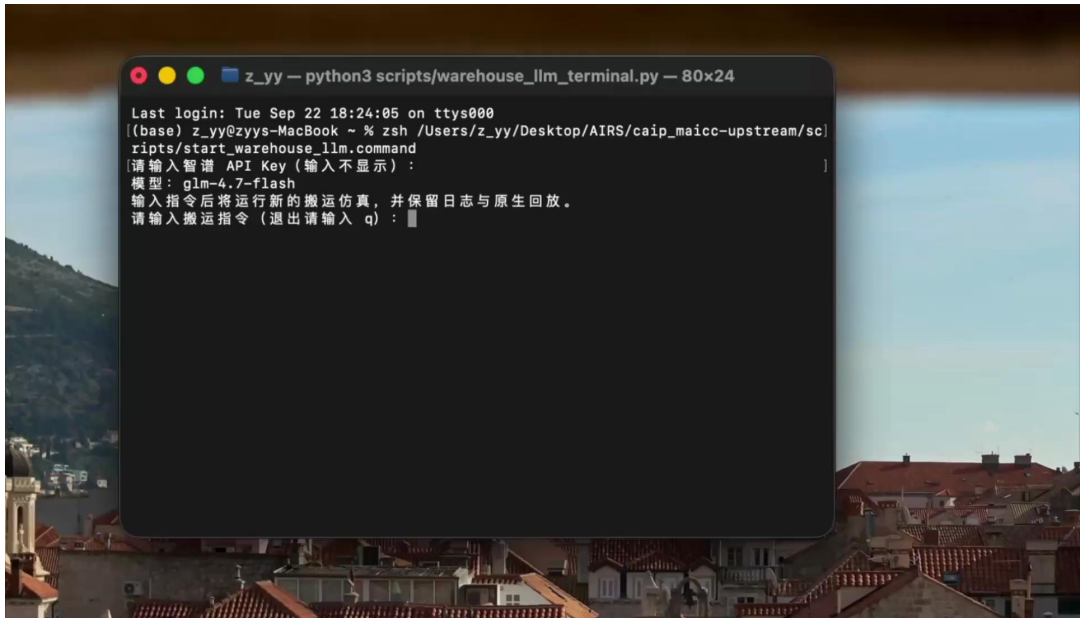
Validation 2: Instruction Understanding & Task Execution

Agents can understand human instructions, autonomously plan and coordinate their actions, and execute tasks to achieve the specified goals.

Example Instructions

- Basic Task: “搬运10个货物，在左下角形成字母A”



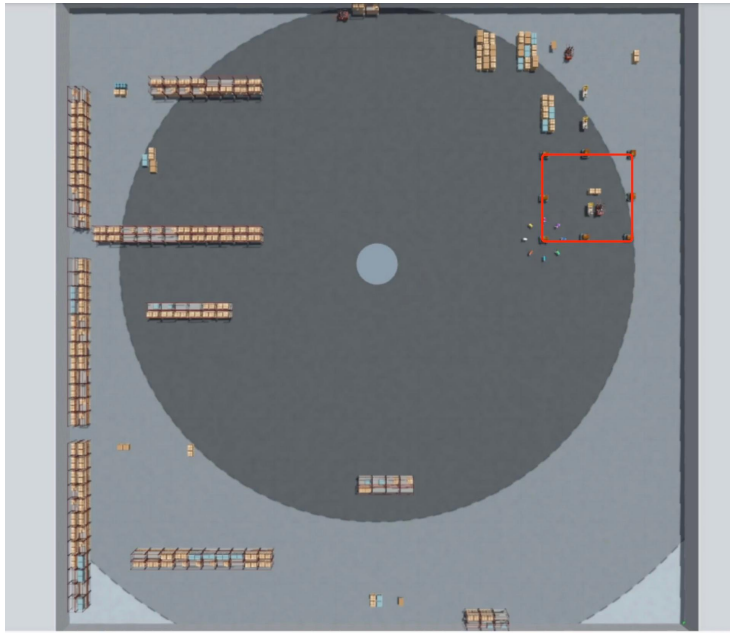


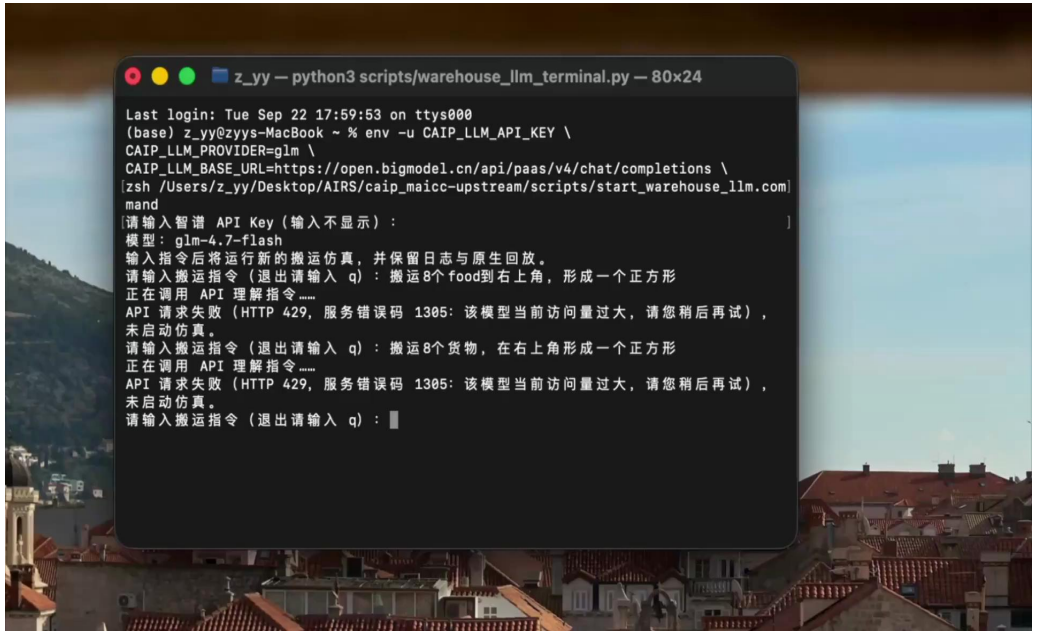
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- Similar Task: “搬运8个货物，在右上角形成一个正方形”



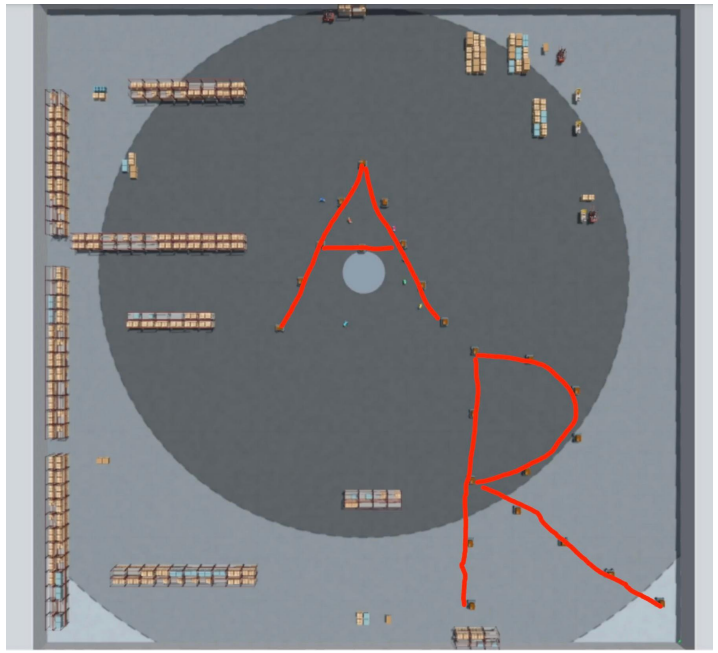


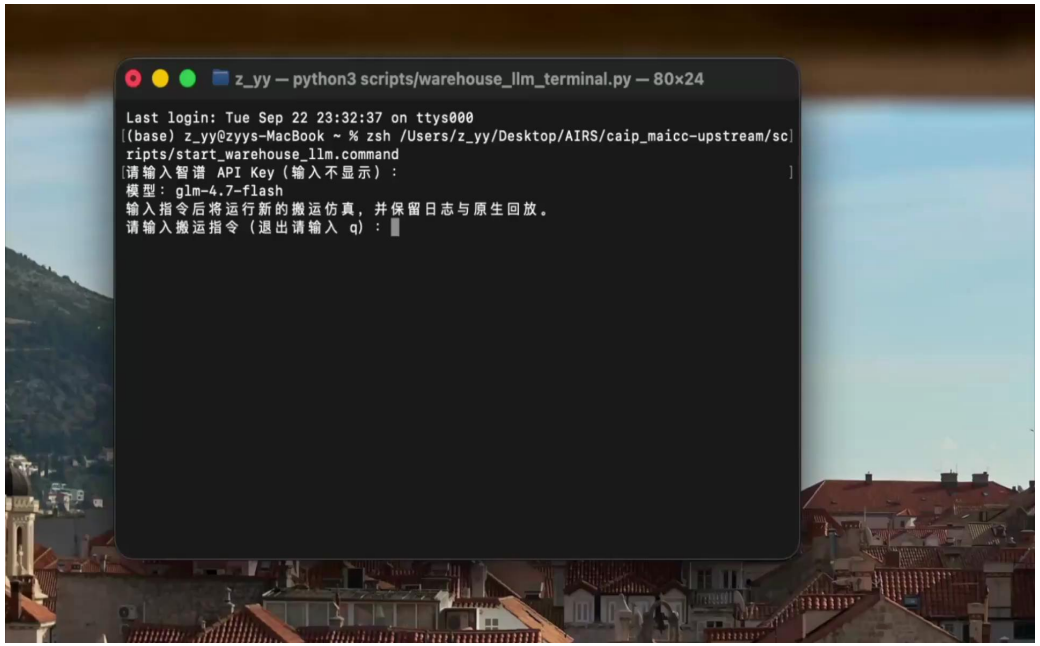
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Example Instructions

- Basic Task: “搬运10个货物，在左下角形成字母A”
- Similar Task: “搬运8个货物，在右上角形成一个正方形”
- Complex Task: “搬运23个货物，用10个在中间形成字母A, 用13个在右下形成字母R”





From Validation to Scalable Multi–Agent Intelligence

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01 VALIDATION 1 — WEAK-NETWORK RESILIENCE

Multi-Agent Coordination Remains Effective Under Weak Communication

- CAIP keeps task execution coordinated as communication degrades or fails.
- Task allocation, adaptive planning, and relaying can operate over imperfect 5G/6G connectivity.

NETWORK FOUNDATION — **Supports wider-area deployment without assuming continuous connectivity.**

02 VALIDATION 2 — INSTRUCTION-DRIVEN EXECUTION

Natural-Language Instructions Can Drive Coordinated Multi-Agent Execution

- Agents interpret user goals, then autonomously plan, coordinate actions, and execute as a team.
- Basic → similar → complex tasks show flexibility beyond one predefined task script.

INTELLIGENCE FOUNDATION — **Toward open-form user instructions driving multi-agent collaboration.**

Representative Application Space

- **Resource Transport**
  Dynamic pickup, teaming, and delivery
- **Disaster Search & Rescue**
  Search, personnel rescue, and supply relay
- **Inspection & Field Operations**
  Distributed inspection, maintenance, and urban services

Resilient 5G/6G coordination + instruction understanding → scalable, intelligent multi-agent collaboration



AIRSEAI Evolution Summary

From execution capability to continuous iteration, and then to system-level coordination

AIRSEAI 1.0

Execution

1.0

How does one robot complete a task?

Task Planning
Navigation / Perception / Grasping
Feedback and Replanning

AIRSEAI 2.0

Engineering Loop

2.0

How does a system continuously improve?

Model Zoo
Capability-Boundary Evaluation
Data Loop

AIRSEAI 3.0

Collective Adaptation

3.0

How can multiple agents coordinate reliably?

CAIP Coordination under Weak Communication
Safety and Scaling

AIRSEAI evolves not by changing robot form factors, but by moving from single-robot execution to engineering loops and then to collective multi-robot coordination.

GOSIM

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Thanks!

